10 Image Segmentation

- Image segmentation divides an image into regions that are connected and have some similarity within the region and some difference between adjacent regions.
- The goal is usually to find individual objects in an image.
- There are fundamentally two kinds of approaches to segmentation: discontinuity and similarity.
  - Partition an image based on abrupt changes in intensity, color or texture but especially sudden changes in intensity along a boundary line, which is called an edge.
  - Partition an image into regions that are similar according to set of predefined criteria

Region-Based Segmentation

- Region-based segmentation is based on the connectivity of similar pixels in a region
  - Each region must be uniform
  - Connectivity of the pixels within the region is very important
- There are two main approaches to region-based segmentation: region growing and region splitting

Basic Formulation

- Let R represent the entire image region.
- Region based Segmentation is a process that partitions R into subregions, $R_1, R_2, \ldots, R_n$, such that

(a) $\bigcup_{i=1}^{n} R_i = R$

(b) R_i is a connected region, $i = 1,2,\ldots, n$

(c) $R_i \cap R_j = \phi$ for all i and j, $i \neq j$

(d) $P(R_i)$ = TRUE for $i = 1,2,\ldots, n$

(e) $P(R_i \cup R_j) =$ FALSE for any adjacent regions R_i and R_j

where $P(R_k)$: a logical predicate defined over the points in set R_k

For example: $P(R_k)$=TRUE if all pixels in R_k have the same gray level

Region Growing Algorithm

- It is a process of grouping the pixels or subregions to get a bigger region present in an image
- **Selection of the initial seed:** Initial seed that represent the ROI should be given typically by the user. Can be chosen automatically. The seeds can be either single or multiple
- **Seed growing criteria:** Similarity criterion denotes the minimum difference in the grey levels or the average of the set of pixels. Thus, the initial seed 'grows' by adding the neighbors if they share the same properties as the initial seed
- **Terminate process:** If further growing is not possible then terminate region growing process

Region Splitting

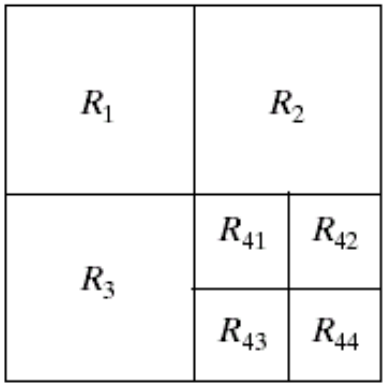
- Region splitting is the opposite of region growing
- Algorithm:
  - First there is a large region (possible the entire image)
  - Then a predicate (measurement) is used to determine if the region is uniform
  - If not, then the method requires that the region be split into two regions
  - Then each of these two regions is independently tested by the predicate (measurement)
  - This procedure continues until all resulting regions are uniform

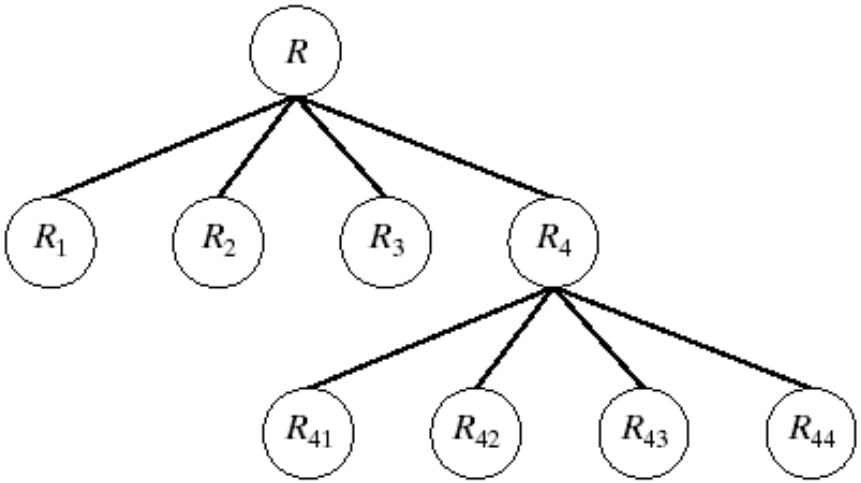
Region Splitting

- The main problem with region splitting is determining where to split a region
- One method to divide a region is to use a quadtree structure
- Quadtree: a tree in which nodes have exactly four descendants

a b

FIGURE 10.42
(a) Partitioned image.
(b) Corresponding quadtree.





Region Merging

- The merge procedure:
  - Merge any adjacent regions R_j and R_k for which $P(R_j \cup R_k)$ = TRUE (the quadtree structure may not be preserved)
  - Stop when no further merging or splitting is possible